

Task 1: Testing Connectivity Between Laptop and Phone

- Connect the laptop and friend's phone to the **same Wi-Fi network**.
- On the laptop, open **Command Prompt**.
- Type `ipconfig` and find the laptop's **IPv4 Address**.
- Find the phone's **IP address** from its Wi-Fi network details.
- On the laptop, type:
`ping [phone IP address]`
- Press **Enter** and observe the results.
- If you receive **Reply from...**, the devices can communicate with each other.
- If you receive **Request timed out**, the laptop cannot communicate with the phone.

Task 2: Testing Network Path Using Tracert

- Open **Command Prompt**.
- Type: tracert www.youtube.com
- Press **Enter**.
- Wait for the command to finish.
- Count the number of **hops** shown in the output.
- Note down the total number of hops in your assignment.
- Take a screenshot of the command output for your report.

Task 3: Testing Connectivity After Disconnecting a Device

- Disconnect the phone from the Wi-Fi network.
- Open **Command Prompt** on the laptop.
- Type ping [phone IP address].
- Press **Enter**.
- Observe the result shown by the ping command.
- Note down the error message or result received.
- Take a screenshot of the output for the report.

Task 4: Listing Devices Using ARP

1. Open **Command Prompt** on your laptop.
2. Type `arp -a`.
3. Press **Enter**.
4. A list of IP addresses and MAC addresses will be displayed.
5. Note down the IP addresses shown in the ARP table.
6. Identify which IP address belongs to which device or person.
7. Create a simple table with the device name, IP address, and MAC address.

Device / Person	IP Address	MAC Address
My Router / Gateway	192.168.29.1	14-ae-85-ea-27-70
Device 1	192.168.29.35	90-32-4b-0b-0d-9d
Device 2	192.168.29.52	00-df-07-fb-b5-ed
Device 3	192.168.29.92	98-da-c4-43-0f-4f
Device 4	192.168.29.124	b4-2e-99-76-77-5e
Device 5	192.168.29.129	38-2c-4a-bd-f7-03

Task 5: One-Page Network Testing Report

I connected my laptop and a friend's phone to the same Wi-Fi network. I used the `ping` command in Command Prompt to test whether the two devices could communicate with each other. The ping results showed whether the phone was reachable from my laptop.

Next, I used the `tracert www.youtube.com` command to check the path from my laptop to YouTube. The command displayed different network hops between my device and the destination. I recorded the number of hops shown in the output.

After that, I disconnected the phone from the Wi-Fi network and tried to ping it again. The ping command showed an error because the device was no longer connected to the network.

Finally, I used the `arp -a` command to view the ARP table. It displayed the IP addresses and MAC addresses of devices known to my laptop. I recorded the relevant devices in a table and identified them where possible.

Overall, these tests helped me understand how **ping**, **tracert**, and **ARP** can be used to test and observe network connectivity.

